Preview of Question Paperset -- PAPERSET I

(PAPERSET I)
Basic Mathematics - I

Mapped Subjects:

[Max Marks: 70]

Note: Attempt Questions from all sections as directed.

Section A - Attempt any two questions out of Four. Each question carries 7.50 marks. [15 Marks]

Question No: 1

Discuss the concept of simple and compound interest. Also, differentiate between them.

Question No: 2

Explain the procedure of finding inverse of a matrix.

Question No: 3

Define discontinuous functions with example.

Question No: 4

Explain the concept of integration by parts with example.

Section B - Compulsory Questions. Each question carries 7.50 marks. [15 Marks]

Let A be the set of first two natural numbers and B be the set {-1,0}.

Question No: 1

Find the Cartesian product $A \times B$.

Question No: 2

Find the power set of the set A. What is the relationship between the number of elements in the power set of A and the number of elements in set A?

Section C - Compulsory Questions. Each question carries 2.00 marks. [40 Marks]

Question No: 1

A person incurs a loss of 5% by selling a watch for ₹1140. At what price should the watch be sold to earn 5% profit.

Option: 1

₹1200

Option: 2

₹1230

Option: 3

₹1260

Option: 4

₹1290

Question No: 2

₹800 becomes ₹956 in 3 years at a certain rate of simple interest. If the rate of interest is increased by 4%, what amount will ₹800 become in 3 years?

Option: 1

₹1052

Option: 2

₹1152

Option: 3

₹1252

Option: 4

₹1352

Question No: 3

Capital investment appraisal refers to:

Option: 1

Making sure that we have enough money to buy assets

Option: 2

Making sure that assets pay for themselves as soon as possible

Option: 3

Making sure that we have enough assets to operate

Option: 4

Evaluating the costs and benefits of proposed investments in operating assets

Question No: 4

Simplify $12\sqrt{5} - \sqrt{125}$

Option: 1

$12\sqrt{5}$

Option: 2

$15\sqrt{5}$

Option: 3

$17\sqrt{5}$

Option: 4

$19\sqrt{5}$

Question No: 5

Rationalizing factor of the surd $\sqrt[4]{5}$

Option: 1

$\sqrt[4]{5}$

Option: 2

$\sqrt[4]{75}$

Option: 3

$\sqrt[4]{125}$

Option: 4

$\sqrt[4]{25}$

Question No: 6

If x = $7 - 4\sqrt{3}$ find the value of $\sqrt{x} + \frac{1}{\sqrt{x}}$.

Option: 1

1

Option: 2

2

Option: 3

3

Option: 4

4

Question No: 7

If x = $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{\sqrt{3}+\sqrt{2}}}$ = y, then find$(x + y)^2$.

Option: 1

0

Option: 2

10

Option: 3

100

Option: 4

1000

Question No: 8

Find the sum of first 25 terms of the following series

10, 15, 20, 25, …

Option: 1

1050

Option: 2

1575

Option: 3

1750

Option: 4

1675

Question No: 9

If a, b and c are in a G.P then a+b, 2b, b+c are in

Option: 1

AP

Option: 2

GP

Option: 3

HP

Option: 4

None of these

Question No: 10

A matrix consisting of m rows and n columns, where $m = n$,is called _____ matrix.

Option: 1

Circular

Option: 2

Triangular

Option: 3

Square

Option: 4

Rectangular

Question No: 11
Find the time required to earn Rs. 500 as simple interest on the principal of Rs. 1000 at the rate of interest 5%.

Option: 1

7 years

Option: 2

8 years

Option: 3

2 years

Option: 4

10 years

Question No: 12

To find the compound amount of ₹ 5000 for 3 years at 8% per annum converted semi-annually by the formula “A = P(1+r/100)^n”, you have to divide the rate of interest by ____.

(a) Annually

Option: 1

2

Option: 2

3

Option: 3

4

Option: 4

None

Question No: 13

Formula for finding “principal”, if rate of interest, time period and simple interest is known.

Option: 1

P = (S.I. * 100)/ (R * T)

Option: 2

P = (S.I. * 100) * (R / T)

Option: 3

P = (S.I. * R)/ (R * 100)

Option: 4

P = (S.I. * T)/ (R / 100)

Question No: 14

A and B being non zero matrices, k and g being scalars, which of the following is a property of scalar multiplication?

Option: 1

(k + g)A = kA + gA

Option: 2

(ABk) = (kA)B = A(k)B

Option: 3

(kA + B) = kA + kB

Option: 4

All of these

Question No: 15

If transpose of a matrix A is equal to –A, then A is a _____

Option: 1

Diagonal matrix

Option: 2

Upper triangular matrix

Option: 3

Skew-symmetric matrix

Option: 4

Symmetric matrix

Question No: 16

Identify the formula for finding inverse of a matrix A.

Option: 1

A^(-1) = 1/ det(A) * (Minors matrix)^T

Option: 2

A^(-1) = 1/ det(A) * (Cofactors matrix)^T

Option: 3

A^(-1) = 1/ det(A) * (Adjugate)^T

Option: 4

All of these

Question No: 17

Depending upon the requirement, which type of operation amongst the following can be called as elementary row operations in order to bring a matrix to row-echelon form?

Option: 1

R_1 ↔ R_2

Option: 2

R_3 → R_3 + 2R_2

Option: 3

R_2 → R_2 + R_1

Option: 4

All of these

Question No: 18

Two matrices A and B can be added or subtracted together, if and only if _____.

Option: 1

Number of rows in matrix A = Number of columns in matrix B

Option: 2

Number of columns in matrix A = Number of rows in matrix B

Option: 3

Order of matrix A = Order of matrix B

Option: 4

First row first column entry of matrix A = Second row second column entry of matrix B

Question No: 19

If determinant value of a matrix is zero, then it is called _____.

Option: 1

Zero matrix

Option: 2

Identity matrix

Option: 3

Singular matrix

Option: 4

Non – Singular matrix

Question No: 20

Mathematically, what is a differential?

Option: 1

A gear box on the back end of your car

Option: 2

A word used a lot on a popular medical television series

Option: 3

A method of directly relating how changes in an independent variable affect changes in a dependent variable

Option: 4

A method of directly relating how changes in a dependent variable affect changes in an independent variable